The Resolution Geometry Atlas

Globalization as a Principal Bundle: Čech Obstructions, Holonomy, and a Falsification Battery

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Abstract

This paper gives the global theory of resolution geometry. A resolution geometry assigns to an observation chart a metric, a resolved sector, a null sector, and a covariance with a resolved–null coupling; the local theory classifies such charts up to admissible isomorphism. Here we assemble local charts into a global object and identify exactly what obstructs the assembly. The result is that the correct upper structure is a *principal bundle* with structure group the chart automorphism group $\mathrm{Aut}(\mathcal{G})$—the coupling-stabilizer group of the local theory—and that globalization is governed by a two-tier obstruction, both levels of which are classical. *Level A* (local admissibility): a candidate transition is admissible iff its sector- and covariance-transport defects vanish, which holds iff it lies in $\mathrm{Aut}(\mathcal{G})$. *Level B* (gluing): admissible transitions define a coherent atlas iff they satisfy the Čech 1-cocycle condition on triple overlaps; the atlas is then a principal $\mathrm{Aut}(\mathcal{G})$-bundle, and its existence class lives in the nonabelian cohomology $H^1(\mathcal{U}, \mathrm{Aut}(\mathcal{G}))$. *Holonomy* is the obstruction not to gluing but to *trivialization*: admissible but nontrivial holonomy is a valid global geometry, and by Ambrose–Singer is generated by the curvature of the projection connection. The decisive structural point is that the local *complete* invariant makes edge-admissibility *decidable* from local data alone, reducing globalization from a formless question to a bundle-classification problem with explicit input. A ten-test synthetic falsification battery confirms every distinction—pairwise admissibility passing while the cocycle fails, nontrivial holonomy remaining admissible, and rank-changing and singular-support charts appearing as genuine boundary cases; the load-bearing tests are independently reproduced. We state precisely what locks (the bundle framework and the reduction, all established mathematics) and what remains open (the full computation of H^1 and the admissible holonomy groups).

Contents

1 Introduction

The local theory of resolution geometry is settled: each observation chart carries a geometry $\mathcal{G} = (O, g; R, \Sigma)$, and charts are classified up to admissible isomorphism by a complete invariant (the block spectra together with the coupling in the block eigenbases, modulo a discrete gauge). The open frontier is *global*: when do local charts, with candidate transition maps on their overlaps, assemble into a single coherent global geometry, and what obstructs it?

This paper answers the structural form of that question. The answer is not new mathematics but the correct identification of established mathematics: the global object is a *principal bundle* whose structure group is the local automorphism group, and the obstructions are exactly those of bundle theory—the Čech cocycle class and the holonomy. What the resolution-geometry setting contributes is that its local invariant is *complete*, so that the usually intractable question "can these charts be glued?" becomes decidable from local data, and globalization reduces to a bundle-classification problem with explicit input.

We are careful to separate three objects that are easily conflated:

(A) the *local admissibility defect* of a single candidate transition—an *admissibility* failure;

(B) the *Čech cocycle defect* on triple overlaps—a *gluing* failure;

(C) the *holonomy* around loops—*not* a failure, but the class of a nontrivial global geometry.

Conflating (C) with (A) or (B) is the standard error; keeping them separate is what makes the theory correct.

Non-claim 1.1 (What is and is not achieved)*.* We establish the upper structure (principal $\mathrm{Aut}(\mathcal{G})$-bundle) and the two-tier obstruction, and prove the globalization criterion; all of this is classical bundle theory instantiated on the resolution-geometry structure group. We do *not* compute the full nonabelian cohomology $H^1(\mathcal{U}, \mathrm{Aut}(\mathcal{G}))$ or classify the admissible holonomy groups in general; those remain open, though now as well-posed standard problems rather than a foundational gap.

2 Local data and the structure group

We recall only what the global theory needs. A chart carries $\mathcal{G} = (O, g; R, \Sigma)$ with $R = \mathrm{Im}(J)$, $N = R^{\perp_g}$, and, in a g-orthonormal adapted frame, $\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix}$. Its *similarity type* is the complete local invariant

$$\tau(\mathcal{G}) = \big(\dim O,\ \dim R,\ \mathrm{spec}\,\Sigma_R,\ \mathrm{spec}\,\Sigma_N,\ \sigma(\Sigma_R^{-1/2} C\, \Sigma_N^{-1/2}),\ \mathrm{Aut}(\mathcal{G})\big),$$

where $\sigma(\cdot)$ denotes the canonical-correlation (singular-value) spectrum of the whitened coupling. The automorphism group is the coupling stabilizer

$$\mathrm{Aut}(\mathcal{G}) = \{\, (A, B) : A\Sigma_R A^\top = \Sigma_R,\ B\Sigma_N B^\top = \Sigma_N,\ A\,C\,B^\top = C \,\}.$$

Definition 2.1 (RG structure group)**.** For a fixed similarity type τ, the resolution-geometry structure group is $G_{\mathrm{RG}}(\tau) := \mathrm{Aut}(\mathcal{G})$, the coupling-stabilizer group above.

Lemma 2.2 (Edge existence is decided by the local invariant)**.** *Two charts admit an admissible transition between them if and only if they have the same similarity type τ. When they do, the set of admissible transitions is a torsor under $G_{\mathrm{RG}}(\tau)$.*

Proof. An admissible transition is an isomorphism of resolution geometries; two charts are isomorphic iff their complete invariants agree (local classification), which is the equality of τ. Given one admissible transition L_0, any other is L_0 composed with an automorphism of the target, so the admissible set is a torsor under $\mathrm{Aut}(\mathcal{G}) = G_{\mathrm{RG}}(\tau)$. □

Lemma 2.2 is the engine of the whole global theory: it makes the existence of an admissible edge a *decidable* comparison of local invariants, and identifies the structure group as the freedom in the choice of edge.

3 Level A: local admissibility

Definition 3.1 (Local transport defect)**.** For a candidate transition $L_{\alpha\beta} : O_\beta \to O_\alpha$ with sector projectors P_R, P_N and covariances $\Sigma_\bullet$, define

$$E^R_{\alpha\beta} = L_{\alpha\beta}P_{R,\beta} - P_{R,\alpha}L_{\alpha\beta}, \quad E^N_{\alpha\beta} = L_{\alpha\beta}P_{N,\beta} - P_{N,\alpha}L_{\alpha\beta}, \quad E^\Sigma_{\alpha\beta} = L_{\alpha\beta}\Sigma_\beta L^\top_{\alpha\beta} - \Sigma_\alpha,$$

and the local defect $\Omega^{\mathrm{loc}}_{\alpha\beta} = (E^R_{\alpha\beta}, E^N_{\alpha\beta}, E^\Sigma_{\alpha\beta})$.

Theorem 3.2 (Admissible transitions are exactly Aut-valued)**.** *$\Omega^{\mathrm{loc}}_{\alpha\beta} = 0$ if and only if $L_{\alpha\beta}$ maps $R_\beta \to R_\alpha$, $N_\beta \to N_\alpha$, and pushes Σ_β to Σ_α—that is, iff $L_{\alpha\beta}$ is an isomorphism of resolution geometries. When $\mathcal{G}_\alpha = \mathcal{G}_\beta$ this is exactly membership in $G_{\mathrm{RG}}(\tau)$.*

Proof. $E^R = E^N = 0$ is intertwining of the projectors, i.e. sector preservation; $E^\Sigma = 0$ is covariance push-forward preservation (Proposition on covariance transport, foundational paper). Together these are the definition of an admissible isomorphism; for equal charts they define the automorphism group. □

Remark 3.3 (The defect is not itself Lie-algebra valued). Ω^{loc} lives in a product of operator spaces, not in $\mathfrak{aut}(\mathcal{G})$. Only after $\Omega^{\mathrm{loc}} = 0$ does the transition lie in the structure group; the failure is measured in the ambient operator space, and admissibility is the vanishing locus, not a subalgebra condition.

4 Level B: Čech gluing and the bundle

Suppose every candidate transition is admissible ($\Omega^{\mathrm{loc}} = 0$), so all $L_{\alpha\beta} \in G_{\mathrm{RG}}(\tau)$. Coherence still requires triple- overlap consistency.

Definition 4.1 (Čech cocycle defect)**.** On a triple overlap $U_\alpha \cap U_\beta \cap U_\gamma$,

$$\Omega^{\check{C}}_{\alpha\beta\gamma} = L_{\alpha\beta}L_{\beta\gamma}L_{\gamma\alpha}.$$

The atlas is *coherent* if $\Omega^{\check{C}}_{\alpha\beta\gamma} = I$ on every triple overlap (equivalently $L_{\alpha\beta}L_{\beta\gamma} = L_{\alpha\gamma}$).

Theorem 4.2 (Static globalization criterion)**.** *Let $\{U_\alpha\}$ be a finite cover with local charts $\mathcal{G}_\alpha$ of a fixed similarity type τ and candidate transitions $L_{\alpha\beta}$. These data define a global resolution geometry—a principal $G_{\mathrm{RG}}(\tau)$-bundle with fiber the chart type—if and only if*

1. *$\Omega^{\mathrm{loc}}_{\alpha\beta} = 0$ on every overlap (Level A), and*
2. *$\Omega^{\check{C}}_{\alpha\beta\gamma} = I$ on every triple overlap (Level B).*

Equivalently, the accepted transitions form a Čech 1-cocycle valued in $G_{\mathrm{RG}}(\tau)$, and the global object is the bundle reconstructed from that cocycle. Its isomorphism class is an element of the nonabelian cohomology $H^1(\mathcal{U}, G_{\mathrm{RG}}(\tau))$.

Proof. By Theorem 3.2 condition (1) places every transition in the structure group. Condition (2) is the 1-cocycle condition. A family of structure-group-valued transitions satisfying the cocycle condition is exactly the reconstruction data of a principal bundle (Steenrod's clutching construction), and two such families define isomorphic bundles iff they differ by a coboundary $L_{\alpha\beta} \mapsto h_\alpha L_{\alpha\beta} h_\beta^{-1}$; the equivalence classes are $H^1(\mathcal{U}, G_{\mathrm{RG}}(\tau))$ [1, 2]. $\square$

Corollary 4.3 (Completeness for globalization)**.** *Local similarity type alone does not determine the global geometry; the data are*

$$\textit{static globalization data} = (\textit{local type } \tau) \ + \ (\textit{Čech cocycle class in } H^1(\mathcal{U}, G_{\mathrm{RG}}(\tau))).$$

The first factor is decidable locally (Lemma 2.2); the second is the global content.

Remark 4.4 (Nonabelian caveat). $G_{\mathrm{RG}}(\tau)$ is generally nonabelian, so $H^1(\mathcal{U}, G_{\mathrm{RG}})$ is a pointed set, not a group; its computation is the hard, open part (Section 9). The criterion of Theorem 4.2 is nonetheless exact and checkable on any finite atlas.

5 Holonomy: obstruction to trivialization, not to gluing

Assume a coherent atlas (Theorem 4.2). Transport around a loop $\gamma = (\alpha_0, \alpha_1, \ldots, \alpha_k = \alpha_0)$ gives the holonomy

$$\mathrm{Hol}_\gamma = L_{\alpha_0\alpha_1} L_{\alpha_1\alpha_2} \cdots L_{\alpha_{k-1}\alpha_0} \ \in \ G_{\mathrm{RG}}(\tau).$$

Proposition 5.1 (Trichotomy of holonomy)**.** *For a coherent atlas,* $\mathrm{Hol}_\gamma \in G_{\mathrm{RG}}(\tau)$ *always, and:*

- $\mathrm{Hol}_\gamma = I$ *for all* γ*: the bundle is trivial (a global trivialization exists);*
- $\mathrm{Hol}_\gamma \in G_{\mathrm{RG}}(\tau)$, $\mathrm{Hol}_\gamma \neq I$ *for some* γ*: a valid global geometry with nontrivial bundle class—* no *global trivialization, but no failure;*
- *a candidate loop product leaving* $G_{\mathrm{RG}}(\tau)$*: the atlas was not coherent (some edge was inadmissible), i.e. a Level-A/B failure, not a holonomy.*

By the Ambrose–Singer theorem the holonomy group is generated by the curvature of the projection connection on the resolved subbundle [3, 2].

Remark 5.2. The essential correction over naive treatments: nontrivial holonomy is the signature of a nontrivial global geometry, exactly as a nontrivial vector bundle has nontrivial transition holonomy without any inconsistency. It is the obstruction to trivialization, not to existence.

6 The obstruction ladder

The three levels form a strict hierarchy, each meaningful only once the previous vanishes.

$$\begin{tikzcd} \boxed{\begin{array}{c}\text{Level A: } \Omega^{\mathrm{loc}} \\ \text{admissibility}\end{array}} \arrow[r, "= 0"] & \boxed{\begin{array}{c}\text{Level B: } \Omega^{\check{C}} \\ \text{gluing (cocycle)}\end{array}} \arrow[r, "= I"] & \boxed{\begin{array}{c}\text{Holonomy } \mathrm{Hol}_\gamma \\ \text{bundle class}\end{array}} \end{tikzcd}$$

Level A is decidable from local invariants (Lemma 2.2); Level B is the Čech class (Theorem 4.2); the holonomy classifies the global geometry once A and B are satisfied (Proposition 5.1). This is the sense in which globalization is *reduced*: a formless "can it be assembled?" becomes a decidable local test, then a cocycle condition, then a bundle class.

7 Boundary cases

Two situations fall outside the fixed-type bundle picture and are genuine boundaries, not failures.

Proposition 7.1 (Rank change requires stratification)**.** *If two charts have different resolved dimension (*rank J *differs), no fixed-type admissible transition exists (Lemma 2.2); the charts belong to different strata, and a global object requires a* stratified *atlas gluing bundles of different types along their boundary.*

Proposition 7.2 (Singular support boundary)**.** *When Σ is singular, the generalized (pseudo-inverse) response is valid iff $R \subseteq \operatorname{range}(\Sigma)$. If R meets $\ker(\Sigma)$, the resolved sector contains a deterministic direction and the whitening/isometry picture is replaced by the support-restricted one; such charts glue only along their common support.*

These are the correct edges of the theory; their full stratified/singular development is open (Section 9).

8 Falsification battery

To confirm that the three levels are genuinely independent obstructions and not artifacts of a single scalar, we ran a synthetic battery on a base chart ($\dim O = 5$, $\dim R = 3$, $\dim N = 2$; base canonical correlations $(0.1530, 0.0737)$). Each test targets one mechanism. The load-bearing cases (base invariants; the Fisher/raw-block false positive; the cocycle-vs-holonomy distinction) were independently reproduced.

Test	Outcome	Mechanism	Key value
B1 identity atlas	pass	trivial coherent atlas	max err 0
B2 admissible gauge	pass	decoupled rotation is in Aut	cov err 10^{-16}
B3 coupling stratum	obstruction	canonical-correlation mismatch	9.6×10^{-2}
B4 Fisher false positive	obstruction	same Schur/Fisher, different raw blocks	c.c. err 0
B5 null-sector info	pass	coupling strictly increases information	2.35×10^{-2}
B6 pairwise vs cocycle	obstruction	edges admissible, triple cocycle fails	5.6×10^{-1}
B7 nontrivial holonomy	pass	holonomy in Aut, $\neq I$ (valid)	8.4×10^{-1}
B8 rank change	boundary	stratified atlas required	$3 \to 2$
B9 singular support	boundary	valid iff $R \subseteq \operatorname{range}(\Sigma)$	—
B10 strict vs conformal	boundary	conformal accepts global scale ray	$\lambda = 2.75$

Table 1: Synthetic falsification battery. Rank-only and Fisher-only criteria are insufficient (B4); canonical-coupling strata and Čech cocycles are independent obstructions (B3, B6); nontrivial admissible holonomy is not a failure (B7); rank-changing and singular cases are boundaries (B8–B10).

Remark 8.1 (The decisive tests)*.* B4 exhibits two representations with identical Schur complement, identical Fisher information, and identical canonical correlations but genuinely different raw (Σ_N, C): they are related by an admissible null-sector gauge, so the invariants agree while the raw blocks do not—confirming that the similarity type, not the raw blocks or Fisher, is the correct object (independently reproduced to $\sim 10^{-16}$). B6 exhibits three pairwise-admissible transitions (each in Aut) whose triple product is a nontrivial rotation, so gluing fails though every edge is valid. B7 exhibits a holonomy in Aut but $\neq I$: a valid nontrivial global geometry. Together these separate Levels A, B, and C empirically.

9 What locks, and what remains open

Locked (established mathematics). The upper structure of resolution geometry is a principal $G_{\mathrm{RG}}(\tau)$-bundle (Definition 2.1, Theorem 4.2); admissible transitions are exactly Aut-valued (Theorem 3.2); coherence is the Čech 1-cocycle condition and existence is classified by $H^1(\mathcal{U}, G_{\mathrm{RG}})$ (Theorem 4.2, Corollary 4.3); holonomy is the trivialization obstruction generated by curvature (Proposition 5.1); and edge-admissibility is decidable from the local complete invariant (Lemma 2.2), which is what reduces globalization to a bundle-classification problem with explicit local input.

Open (now well-posed).

1. Computation of the nonabelian cohomology $H^1(\mathcal{U}, G_{\mathrm{RG}}(\tau))$ for structure groups arising from concrete coupling strata.
2. Classification of the admissible holonomy groups (subgroups of $G_{\mathrm{RG}}(\tau)$ realizable as Hol_γ).
3. Stratified atlases across rank-changing boundaries, and singular-support gluing.
4. The connection whose curvature realizes the holonomy (the Ambrose–Singer generator) as an explicit object in the block data.

Non-claims. No claim is made that a coherent global geometry always exists for given local data—its existence is exactly the content of the cocycle condition, which can fail (B6). No claim is made to have computed the global classification; the contribution is the correct framework and the reduction.

10 Conclusion

Globalization of resolution geometry is neither mysterious nor open at the level of structure: it is the assembly of local charts into a principal bundle whose group is the local automorphism (coupling-stabilizer) group, obstructed at two classical levels—local admissibility and Čech coherence—with holonomy classifying the nontrivial but valid global geometries. The one feature that makes the resolution-geometry setting tractable, and that distinguishes it from a generic bundle problem, is that its local invariant is complete: edge-existence is decided by comparing local data, so the global problem collapses to a cocycle class and a holonomy. A synthetic battery confirms the three obstruction levels are independent and behave exactly as the theory predicts. What is established is the framework and the reduction, in established mathematics; what remains is the computation of the resulting cohomology and holonomy classes—a hard but well-posed problem, no longer a foundational void.

References

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